



What is Continuous Delivery (CD)

Continuous Delivery (CD) is a software development practice where code changes are automatically built, tested, and prepared for release to production so that software can be deployed at any time with minimal manual effort.

How Continuous Delivery Works

1. **Developers commit code** to a shared repository.
2. The system automatically:
 - Builds the application
 - Runs automated tests
 - Validates quality checks
3. If everything passes, the system prepares the application for release.
4. A human can deploy it to production with a simple approval (often just a button click).

Deployment could also mean:

- Server Provisioning
- Dependencies
- Conf Changes
- Network/ Firewall rule changes
- Artifact Deploy

Automation Tools:

- Ansible, Puppet, Chef (System Automation)

- Terraform, CFormation (Cloud Info Automation)
- Jenkins, Octopus Deploy (CI/CD Automation)
- HELM Charts
- Code Deploy

Software Testing must be automated as well, these test processes could be:

- Functional
- Load
- Performance
- DataBase
- Network/Security